

Tiga: Autonomous Hybrid Active-Passive Communication for Ambient IoT System

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Abstract—Power consumption continues to pose a fundamental challenge for large-scale IoT tag deployment. While conventional active communication strategies struggle to mitigate peak power demands due to energy-intensive components, passive alternatives achieve microwatt-level consumption but suffer from limited operational range. This paper introduces *Tiga*, an innovative hybrid active-passive communication system for Ambient IoT that autonomously selects optimal transmission modes. Recognizing the ambiguity inherent in the distance-based mode switching boundary, we instead adopt energy level as our boundary criterion. *Tiga* employs passive communication when channel energy suffices and seamlessly transitions to active transmission when required. Our design features a novel microwatt-scale dual-function detector that performs concurrent downlink signal demodulation and RSSI computation. We further propose an adaptive four-phase blind boundary search algorithm capable of efficient execution on any low-power controller. We prototype *Tiga* on a four-layer PCB and evaluate its performance across various environments. Experimental results demonstrate that compared to duty-cycling active communication, *Tiga* reduces dynamic power consumption by $1.85\times$ to $9.42\times$ across various distances from the gateway.

Index Terms—Low-power, Internet of Things, Hybrid communication, Ambient IoT

I. INTRODUCTION

THE Internet of Things (IoT) bridges the physical and digital worlds by embedding tags into ubiquitous physical objects. These IoT tags enable real-time monitoring of environmental changes and object behaviors, facilitating seamless connectivity between the two realms. Such connectivity has found widespread applications across various domains, including human health monitoring [1], asset tracking [2], and environmental human-computer interaction [3]. A critical challenge in IoT tag design lies in minimizing power consumption to ensure sustained operation without frequent, labor-intensive human intervention.

Recent studies have explored strategies such as duty cycling [4], [5], data compression [6], [7], and intermittent energy harvesting [8], [9] to reduce power consumption. However, these mechanisms fail to address the peak power consumption caused by high-power components like carrier generators and amplifiers.

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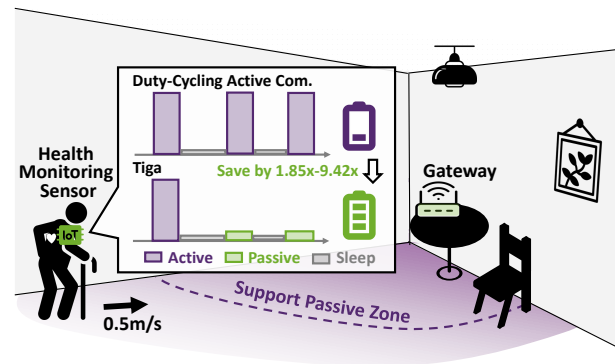


Fig. 1. An illustration of *Tiga* deployment for elderly health care. *Tiga* autonomously determines the operating mode based on RSSI and implements both active and passive communication with duty-cycling.

Ambient IoT (A-IoT) represents the next evolutionary stage of IoT tags. This emerging class of IoT tags is characterized by ultra-low power consumption, minimal maintenance requirements, and cost-effectiveness, leveraging ambient energy harvesting from diverse sources including RF signals, solar/light, motion, and thermal gradients [10]. The technological cornerstone of A-IoT is backscatter communication¹, which utilizes ambient RF signals without dedicated carrier generation, achieving microwatt-level power consumption. However, passive communication faces the challenge of double-path loss, making its performance highly dependent on the location and strength of RF sources. Consequently, reliable communication is typically limited to areas near gateways [11], [12], [13], [14], [15], [16], leaving active communication to play a crucial role in A-IoT applications.

In recent years, standardization organizations such as 3GPP [17], the Bluetooth Special Interest Group [18], and IEEE 802.11 AMP [19] have focused on defining optimal support mechanisms for A-IoT tags. 3GPP categorizes tags into three classes (Device 1, 2a, and 2b) based on application requirements, where Device 1 and 2a support passive communication while Device 2b supports active communication [17]. A single tag can support multiple types simultaneously, making hybrid active-passive architectures a potential solution where tags can dynamically select optimal communication modes. Prior works [20], [21], [22], [23], [24], [25] have investigated mode switching in hybrid systems, though these studies lack real-world prototype evaluations

¹ Backscatter communication is a passive communication technique distinct from active methods. Throughout this paper, we use “passive communication” to refer specifically to backscatter technology.

TABLE I
COMPARISON BETWEEN DIFFERENT SOTAS.

SoTA	Active Support	Passive Support	Energy Harvesting	Hardware Implementation	Power Consumption (Active/Passive)	Mode Switching Strategy
TinySDR[42]	✓	X	X	✓	>500mW / NA	/
MilliMobile[9]	✓	X	✓	✓	>35mW / NA	/
Flync[35]	✓	X	✓	✓	>35mW / NA	/
AirTag[34]	✓	X	X	✓	NA	/
Wiliot IoT Pixels[43]	✓	X	✓	✓	NA	/
FaB[48]	X	✓	X	✓	NA / 176.49mW	/
PLoRa[46]	X	✓	✓	✓	NA / 2.6mW	/
Backscatter-IC[53]	X	✓	✓	✓	NA / 43.2uW(IC)	/
HaiLa BSC2000[56]	X	✓	X	✓	NA / 25uW(IC)	/
Hybrid-IoT[20]	✓	✓	X	X	NA	Assigned
Hybrid-MEC[25]	✓	✓	✓	X	NA	Assigned
Hybrid-WPCN[22]	✓	✓	✓	X	NA	Autonomous
Hybrid-BLE[27]	✓	✓	X	✓	81mW / 32.6mW	Assigned
Braidio[26]	✓	✓	X	✓	129mW / 2mW	Assigned
Tiga	✓	✓	✓	✓	90.12mW / 2.9mW	Autonomous

and operate under idealized assumptions. While [26] and [27] implement prototypes, their mode switching relied on manual assignment. Given that active communication exhibits more robust links[28], it remains unknown whether tags can support double-path loss in passive communication. Ideally, A-IoT tags should autonomously select operational modes based on deployment environments.

In this paper, we present *Tiga*, a hybrid active-passive system that autonomously adapts to the energy conditions of communication links to select optimal communication modes. We observe that the gateway generally satisfies the communication link requirements of the farthest tag by employing high transmission power. This behavior enables tags located closer to the gateway with the opportunity for passive communication. These tags can engage in passive communication within a short range; however, they must generate carriers to sustain the communication link at greater distances. Therefore, based on the distance between the tag and the gateway, there is a physical boundary: passive communication is feasible within this boundary, while active communication becomes necessary beyond it, as shown in Fig. 1.

The key to *Tiga* lies in accurately identifying the mode-switching boundary, which introduces three challenges:

(1) **Physical characterization of boundary.** A naive approach is to directly use the distance between the gateway and the tag as the decision criterion. However, in real-world environments, factors such as multipath effects, obstacles, and interference can render this distance-based boundary unreliable. One deterministic factor for successful backscatter signal reception lies in the energy strength of the signal. Physical distance can be characterized by channel energy loss, while other environmental factors are also reflected in energy variations. IoT tags can measure the signal strength of the received carrier signal, thereby dynamically selecting appropriate operational modes to effectively transform physical boundaries into energy-based boundaries.

(2) **Low-power continuous energy monitoring.** Commercial RF chips, such as the Semtech SX1276 [29], TI CC2541 [30], and MICRF213 [31], designed for active com-

munication, typically offer high-sensitivity energy detection capabilities. However, these chips rely on power-intensive components, including built-in low-noise amplifiers (LNAs), local oscillators, and mixers, which significantly increase power consumption. Even when the RF functionality is disabled, the power consumption remains at approximately 10 mW, posing a substantial challenge for applications requiring continuous energy detection under stringent power constraints. We address this with a microwatt-level dual-function detector that reuses envelope detector outputs to simultaneously demodulate downlink signals and monitor RF signal energy in real time.

(3) **Autonomous boundary discovery.** A-IoT deployments often feature non-line-of-sight conditions and dynamic environments. Structural changes and tag mobility may cause rapid link fluctuations, leading to connectivity loss. If *Tiga* relied on fixed boundary configurations, it would require prior knowledge of deployment scenarios, limiting adaptability. Moreover, when tags are associated with different gateways, previous configurations become invalid. We propose a four-phase blind boundary search algorithm that continuously refines the feasible range by updating upper and lower bounds, ultimately converging to the optimal boundary.

In summary, we make the following contributions:

- To the best of our knowledge, we present for the first time a hybrid active-passive prototype system for A-IoT applications, which can autonomously select the communication mode.
- We design an efficient microwatt-level circuit supporting simultaneous signal demodulation and continuous channel energy detection. For boundary discovery, we develop a four-stage blind search algorithm that refines feasible ranges through bound updates.
- We prototype this hybrid system on a four-layer PCB supporting LoRa PHY with RF/solar energy harvesting. Experiments show *Tiga* reduces dynamic power consumption by $1.85\times$ to $9.42\times$ compared to duty-cycled active communication across various gateway distances.

II. RELATED WORK AND MOTIVATION

Power consumption reduction remains a fundamental challenge for IoT tags, a topic that has been studied for decades. State-of-the-art (SoTA) solutions have extensively explored active, passive, and hybrid communication paradigms, as shown in Table. I.

Active communication systems. [32], [33] use a low-power radio to wake up high-power ones. [4], [5], [34] exploit duty-cycling to shut down the RF radio between transmissions. [6], [7] compress data before transmission to reduce the power consumed by RF radios. [8], [9], [35] intermittently harvest energy to keep the system working. Other works [36], [37], [38], [39], [40], [41] leverage active communication to enable advanced sensing capabilities. TinySDR[42] utilizes the power management module to control the operating state of each component at a granular level to reduce power consumption. Wiliot IoT Pixels[43] designs the first active tag for A-IoT with SoC containing ARM Cortex M0 and BLE RF front-end. However, these methods do not fundamentally solve the high power consumption issue due to power-hungry components during communication.

Passive communication systems. Novel backscatter communication systems enable carrier offloading to constrain power consumption at the microwatt level, categorizing them as passive communication systems. Capitalizing on the inherent advantages of backscatter communication, recent research focuses on enhancing communication performance, such as [13], [44], [14], [45], [46], [47], [48], [49], [50], [51], [52]. Additionally, [53], [54], [55], [56], [57] have advanced ASIC designs, achieving measured communication power consumption at the microwatt level. Despite achieving ultra-low power and cost-effective communication, backscatter technologies still fall short of meeting the demands of real-world IoT applications due to their limited operational range and unreliable communication processes.

Recently, Saiyan [58] has introduced a novel ultra-low-power demodulator for LoRa backscatter systems. By ingeniously leveraging the frequency-amplitude response of SAW filters, Saiyan enables tags to demodulate downlink LoRa chirp signals with microwatt-level power consumption, significantly extending the passive communication range to over 150m. While Saiyan represents an attractive solution for symmetric LoRa backscatter links, *Tiga* aims to address a different challenge: enabling autonomous mode switching within the hybrid active-passive architecture.

Hybrid communication system. Hybrid communication intelligently combines the advantages of both active and passive modes. The system maintains low average power consumption while employing active transmission to counteract double-path loss. Given the substantial energy overhead of active communication, hybrid tags require more generous energy budgets[28].

On one hand, prior works [59], [20], [21], [60], [22] investigate battery-powered solutions where tags derive energy from internal batteries. Such tags may opt for passive communication to mitigate rapid battery depletion, thereby extending overall operational lifetime. On the other hand,

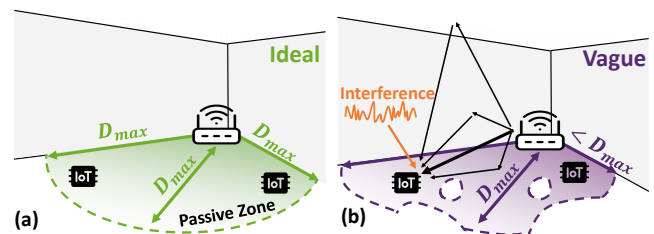


Fig. 2. Physical distance boundary analysis. (a) In ideal case. (b) In real world.

studies [61], [62], [63], [23], [64], [65], [66], [24], [25], [67] explore hybrid systems powered by energy harvesting technologies that extract energy from ambient sources (e.g., RF signals, solar, and mechanical energy). Additionally, [68] employed unmanned aerial vehicles as mobile power beacons, while [69], [70] leveraged reconfigurable intelligent surfaces to maximize energy signal strength. Although these works advance hybrid communication paradigms, none validated their feasibility through prototype implementations. Practical deployment constraints in complex environments would further limit their applicability.

The seminal Braidio system [26] presents an end-to-end hardware platform addressing asymmetric energy budgets, though it exclusively considered mobile tags' battery capacities while ignoring infrastructure-side energy resources. Hybrid-BLE [27] innovatively reuses a single field-effect transistor stage as both a Class-C power amplifier for active transmission and a passive modulator. However, this work focuses solely on RF front-end integration without addressing mode-switching strategies or networking challenges.

Motivation. Driven by A-IoT trends, *Tiga* expects to pioneer the practical deployment of hybrid active-passive communication systems with autonomous mode selection. First, existing hybrid communication studies lack real-world prototype evaluations, typically operating under idealized assumptions. Second, due to the robustness of active links, gateway devices cannot determine when a tag can establish a passive link. Therefore, the hybrid tag must incorporate an autonomous mode-switching mechanism. *Tiga* is designed to enable tags to dynamically select between active and passive modes in real-world deployments, maximizing energy efficiency.

Tiga is well-suited for various application scenarios, such as monitoring the physical health of elderly individuals (e.g., heart rate, blood pressure, sleep patterns) and managing inventory in warehouses. In these scenarios, tags operate within the coverage area of a gateway, allowing them to move closer to the gateway to enable passive communication. However, these applications also require stable, full-range links to maintain continuous connectivity. Therefore, *Tiga* effectively addresses this trade-off by supporting passive communication near gateways and switching to active communication for longer distances.

III. BOUNDARY ANALYSIS

In hybrid active-passive IoT systems, the fundamental operational challenge lies in determining the optimal switching point between modes. To maximize energy efficiency, the tag

should prioritize passive communication whenever the channel conditions permit and switch to active transmission only when necessary. In this section, we analyze the link budget constraints to theoretically derive this operational boundary.

A. Opportunity: Excess Power in Downlink

In a typical star topology, a gateway utilizes a transmit power, denoted as P_{tx} , to ensure connectivity with the farthest node at the cell edge. For an IoT tag i located at a distance d_i from the gateway, the received signal power $P_{rx}(d_i)$ follows the inverse-square law. According to the Friis transmission equation [71], the received power is given by:

$$P_{rx}(d_i) = P_{tx} G_{tx} G_{rx} \left(\frac{\lambda}{4\pi d_i} \right)^2 \quad (1)$$

where G_{tx} and G_{rx} are antenna gains, and λ is the wavelength.

Since the gateway's P_{tx} is dimensioned for the worst-case distance d_{max}^{cell} , any tag located at a closer distance $d_i < d_{max}^{cell}$ receives power significantly higher than the receiver sensitivity. This excess power creates an opportunity: the tag can utilize the strong carrier signal to perform backscatter communication, thereby offloading the power-hungry carrier generation process to the gateway.

B. Constraint: Round-Trip Path Loss

While the downlink active signal decays proportional to d^{-2} (one-way path loss), the passive backscatter uplink suffers from round-trip path loss. The signal travels from the gateway to the tag and reflects back to the gateway. Consequently, the power of the backscattered signal received by the gateway, P_{bs-rx} , is proportional to the product of the forward and backward path losses:

$$P_{bs-rx}(d_i) \propto \frac{1}{d_i^2} \times \frac{1}{d_i^2} = \frac{1}{d_i^4} \quad (2)$$

This drastic d^{-4} attenuation implies that passive communication is only feasible within a limited range, defined as the *Passive Zone*. Outside this zone, the backscattered signal strength falls below the gateway's decoding sensitivity (γ_{gw}), necessitating the use of active transmission (which only suffers d^{-2} loss) to maintain the link.

C. Defining the Optimal Boundary

We define the optimal boundary as the critical state where the backscatter link budget is exactly exhausted. Let η represent the tag's backscatter reflection efficiency. The condition for successful passive communication is

$$P_{bs-rx}(d_i) = \underbrace{P_{rx}(d_i)}_{\text{Tag Reflection}} \cdot \underbrace{\eta \cdot L_{uplink}(d_i)}_{\text{Uplink Loss}} \geq \gamma_{gw} \quad (3)$$

Solving for distance d , we obtain a theoretical maximum distance for passive operation, denoted as D_{max} .

- If $d_i \leq D_{max}$: The tag resides in the *Passive Zone*. It should operate in passive mode to minimize power consumption (μW -level).
- If $d_i > D_{max}$: The tag resides in the *Active Zone*. It must switch to active mode (mW-level) to maintain connectivity.

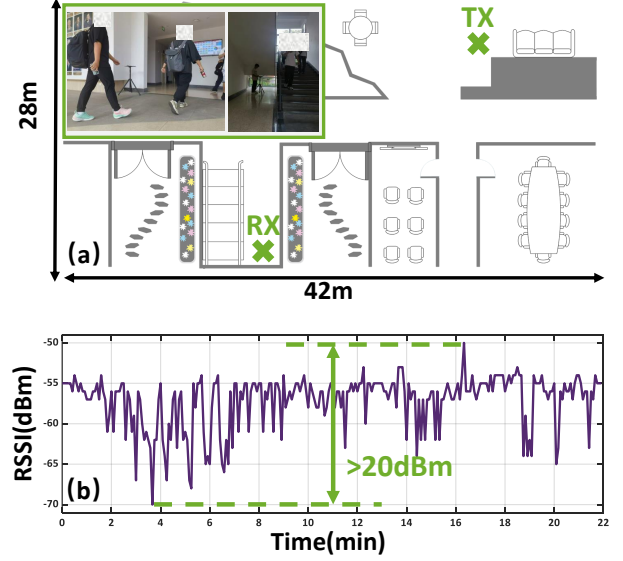


Fig. 3. Temporal Instability. (a) Experiment settings. (b) RSSI fluctuation curve during a 22-minute period.

Therefore, D_{max} represents the physical demarcation line that dictates the optimal mode selection strategies. Ideally, if the channel loss around the gateway is uniform, the optimal boundary is a circle centered on the gateway, as shown in Fig. 2(a).

IV. TIGA DESIGN

A naive approach is to directly derive the distance based on the positional relationship between the gateway and the tag, using D_{max} as the mode selection criterion. However, the stochastic nature of real-world wireless channels renders this physical boundary ambiguous, as illustrated in Fig. 2(b). Furthermore, accurate channel estimation or distance ranging is often prohibitive for IoT tags due to their stringent resource constraints [72]. Consequently, a fundamental challenge arises: *how can a tag autonomously identify the optimal operational boundary and select the appropriate communication mode without incurring excessive overhead?*

In this section, we present a holistic hardware-software co-design to address these challenges. We begin by empirically characterizing the ambiguity of the physical distance boundary, which motivates our transition to an implicit energy boundary. Building on this insight, we detail the implementation of our hybrid active-passive platform designed for A-IoT applications. Subsequently, we propose a dynamic mode selection scheme that enables tags to efficiently search for this energy boundary. Finally, we introduce a protocol to support concurrent access for multiple tags.

A. Real-world vague boundary

In contrast to the deterministic theoretical boundary, the operational boundary in practical deployments is significantly compromised by environmental stochasticity. Factors such as multipath fading, shadowing, and temporal interference obscure the monotonic correlation between physical distance and signal integrity, rendering D_{max} a dynamic variable rather than a static boundary.

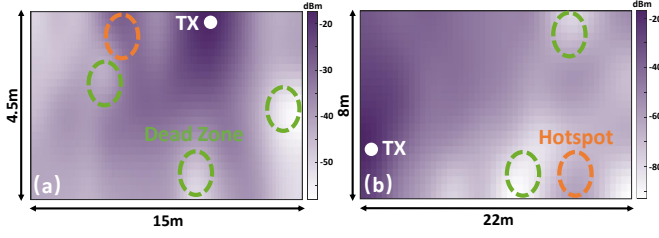


Fig. 4. Spatial Non-uniformity. (a) An office. (b) An express station.

To validate this vagueness, we conduct extensive measurements in typical indoor scenarios.

- **Temporal Instability:** As shown in Fig. 3(a), we place the transmitter and receiver at a fixed location within a school building and monitor the downlink RSSI. The output power of the transmitter is set to 14dBm in 470MHz . As shown in Fig. 3(b), the RSSI fluctuated by over 20dB within a 22-minute window due to environmental dynamics, such as crowd movement. Relying on a fixed distance metric in such a dynamic setting would lead to frequent communication failures.
- **Spatial Non-uniformity:** We fix the transmitter position with an output power of 14dBm and move the receiver to study if the RSSI decreases as the distance increases. We map the spatial distribution of RSSI in an office and an express station (Fig. 4). The results reveal that signal strength does not decay monotonically with distance; instead, multipath effects create “dead zones” near the gateway and “hotspots” at further distances. In these “dead zones”, the useful signal energy would drop below the minimum value required by the receiver sensitivity. Consequently, passive communication fails in these zones, even if the tag is geometrically located well within the theoretical $D_{\max}$.

These empirical observations fundamentally shape *Tiga*'s design philosophy. The temporal instability implies that mode selection cannot be a one-time static configuration but requires continuous adaptation. The spatial non-uniformity confirms that physical distance is a deceptive metric for link reliability.

Therefore, *Tiga* abandons distance estimation and anchors its design on the implicit energy boundary. Since the received signal energy captures the aggregate effect of path loss, fading, and interference, it serves as the reliable ground truth. We define a critical energy boundary, P_{th} , derived from the link budget analysis in Eq. 3. If the sensed downlink energy satisfies $P_{rx} \geq P_{th}$, the channel quality is sufficient to support passive backscatter, regardless of the physical distance. Conversely, if $P_{rx} < P_{th}$, the tag must switch to active transmission to close the link.

By using P_{th} as the decision criterion, *Tiga* effectively bypasses the complexity of distance estimation and adapts naturally to the instantaneous channel conditions.

B. Hybrid active-passive platform

The strategy of exploiting the implicit energy boundary necessitates continuous channel monitoring, which creates a hardware dilemma. Relying on standard active radios for sensing consumes excessive power (e.g., $> 10\text{mW}$), negating

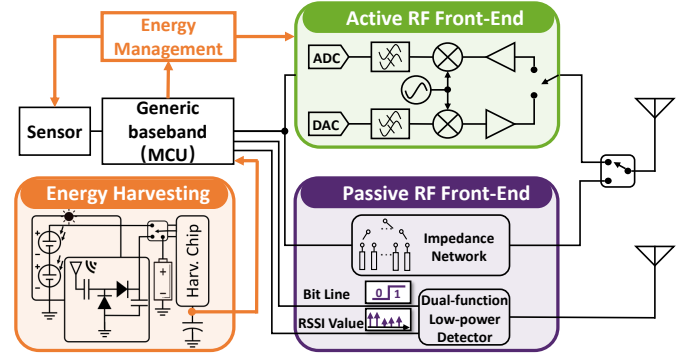


Fig. 5. *Tiga* tag hardware overview.

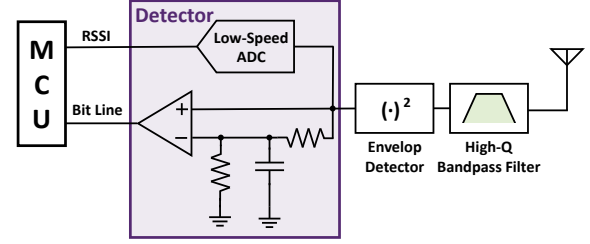


Fig. 6. Dual-function low-power detector circuit design.

the efficiency of passive mode, whereas traditional backscatter tags lack the necessary sensing capabilities. *Tiga* resolves this by introducing a hybrid architecture.

1) **Hardware overview:** The overall design of the *Tiga* tag hardware is illustrated in Fig. 5. The system comprises five main components: (1) **Active RF front end:** This component is primarily used for active communication, incorporating a high-power mixer and power amplifier. (2) **Passive RF front end:** The transmitter (TX) is realized by the baseband signal controlling the RF switch with an impedance network, while the receiver (RX) is implemented using a dual-function low-power detector circuit. Both RF front ends are controlled by the Microcontroller Unit (MCU). (3) **Generic baseband:** Deployed on the MCU, it is responsible for signal generation and control. (4) **Energy management:** *Tiga* tag divides power domains to granularly shut down module power according to its operational mode, thereby reducing static power consumption. (5) **Energy Harvesting:** This module consists of a one-stage rectifier for RF energy harvesting and photovoltaic panel for solar energy harvesting. The output of the harvested energy is connected to the energy management chip.

2) **Dual-function low-power detector:** As discussed in §IV-A, selecting the appropriate operation mode requires measuring the energy. We use RSSI to denote energy values. Although commercial RF chips (e.g., Semtech SX1276[29], TI CC2541[30], and MICRF213 [31]) for active communication generally have the capability to measure RSSI with high sensitivity, we aim to continuously monitor the environment with low power consumption throughout the operating cycle. Moreover, these commercial RF chips contain high-energy-consuming components such as built-in LNAs, local oscillators and mixers, which still cause power consumption to reach the 10mW level.

In the *Tiga* tag, we design a simple yet effective low-power detection circuit that combines RSSI calculation and demod-

ulation, as shown in Fig. 6. We make a trade-off between sensitivity and power consumption in calculating RSSI.

Commercial RF chips have better receiver sensitivity, for example, SX1276 and MICRF213 [31] have automatic gain calibration after amplifiers, which improves the precision of RSSI measurement. Due to the double-path loss, the received signal strength at the tag is much greater than the sensitivity of the gateway receiver [11]. Therefore, we can greatly relax the requirements for tag to calculate RSSI.

Our design multiplexes the output of the envelope detector for both the comparator and the low-speed ADC. The signal detector uses a passive envelope detection circuit with a comparator to demodulate OOK signals. This circuit consumes extremely low power and is widely adopted in the downlink design of backscatter systems [12], [11], [47], [73]. Notably, we choose a high-Q bandpass filter for physical isolation and interference suppression. We use a low-speed ADC to simultaneously perform analog-to-digital conversion of the envelope detector's output to obtain the RSSI. The power consumption of the low-speed ADC can be limited to the $10\mu W$ level.

We argue that the RC circuit and comparator are necessary. Although the comparator is equivalent to a 1-bit ADC, an adaptive threshold circuit is the key to determining the voltage level (i.e., high or low). In this case, a dual-channel ADC is required, which would add additional power consumption.

The outputs of the comparator and the low-speed ADC are connected to the MCU. We implement a moving average filter in software as the low-pass filter, which can be easily reconfigured by setting different window sizes to smooth high-frequency noise.

3) *Generic Baseband*: In *Tiga*, the baseband signal is generated by the MCU and exists in two separate RF front-end paths: (1) The MCU communicates with and controls the RF chip through on-chip peripherals such as SPI [29] and UART [30]. (2) The MCU controls I/O ports to generate baseband signals and frequency shift signals, which are mixed with the carrier via an RF switch. The RF switch connects to an impedance network to achieve changes in the impedance coefficient.

4) *Energy Harvesting Module*: A-IoT tags operate in diverse and complex environments, including indoor, outdoor, and high-temperature scenarios, necessitating versatile energy sources. For instance, in typical indoor settings such as automated warehousing, tags cannot rely on solar or thermal energy and instead depend on RF harvesting. Conversely, in outdoor applications like smart agriculture, tags can harvest energy from ambient sunlight. Therefore, the *Tiga* tag must not be limited to a single energy source but should support extensible energy harvesting interfaces.

As illustrated in Fig. 5, the energy harvesting module incorporates circuits for multiple energy sources. We design the energy management module based on the BQ25570 chip [74], which efficiently regulates energy usage and stores harvested energy in a storage capacitor. The BQ25570 achieves a 93% energy conversion efficiency and supports a peak output current of up to $110mA$, meeting the requirements of *Tiga*'s active module. Additionally, its quiescent current is only $488nA$,

TABLE II
POWER DOMAINS IN *Tiga* TAG.

Component	Power Domain
MCU	V1
Radio	V2
Low-speed ADC	V3
Others (e.g., Sensor)	V4

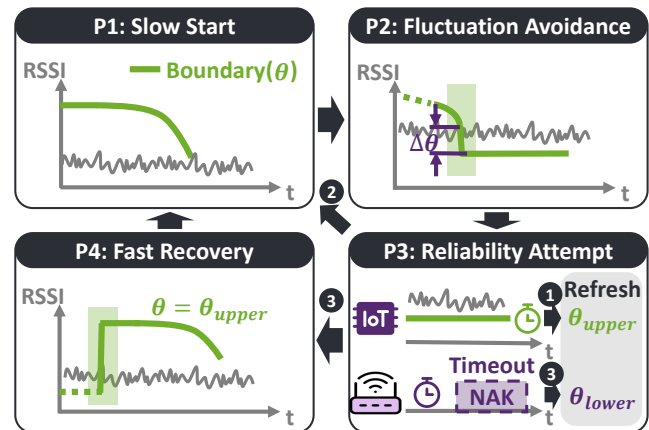


Fig. 7. Four-phase blind boundary search algorithm overview.

minimizing energy waste. We demonstrate *Tiga*'s capability to harvest RF energy and solar energy in the prototype system in §VI-C. Furthermore, we provide a battery interface to enhance *Tiga*'s applicability in battery-powered solutions.

5) *Energy management*: Next, we present the design of the tag's energy management system, which seeks to maximize the system's lifetime when operating on a limited energy. To achieve long life time, we need to duty-cycle the system and allow the MCU to toggle each of the aforementioned blocks *ON* and *OFF* when not in use. Additionally, different components have varying supply voltage requirements, and we aim to provide efficient energy supply while reducing the static power consumption of each block. Ideally, separate controllable voltage regulators for each component would be preferred, but this increases complexity, component count, and cost. Therefore, we balance power control granularity with design complexity and cost by grouping components, as outlined in Table. II.

C. Dynamic mode selection scheme

Tiga tag measures RSSI as a criterion for mode selection, avoiding vague distance boundary in the real world. The hardware design described in §IV-B demonstrates the capability to detect RSSI and perform hybrid active-passive communication. However, when *Tiga* associates with a new cell, the optimal boundary is implicit. To address this challenge, we design a *blind boundary search algorithm*, to search for the boundary automatically. This algorithm operates in four phases, as illustrated in Fig. 7.

Core idea. The primary challenge for the tag lies in determining the minimum received signal energy required to sustain passive communication. To circumvent this, *Tiga* adopts a proactive "trial-and-error" heuristic, where the tag prioritizes transitions to passive mode. Upon confirming link stability,

the tag incrementally decreases the switching boundary to optimize energy conservation. Conversely, if the gateway reports packet loss, the tag infers that the current energy is insufficient for backscatter operation. Consequently, it immediately reverts to the robust active mode and conservatively raises the boundary to prevent future failures. This continuous probing mechanism enables *Tiga* to autonomously converge on the optimal switching boundary.

Specifically, *Tiga* employs a low-power ADC to quantize RF signal energy into digital RSSI values. We denote θ as the energy boundary. If the RSSI exceeds the boundary θ , the tag switches to passive mode; otherwise, it reverts to active mode. The boundary θ serves as the sole criterion for *Tiga* mode selection and is represented numerically after ADC sampling. *Tiga* seeks to determine an optimal θ that falls within the ADC's measurable interval. Consequently, there exists an upper bound θ_{upper} and lower bound θ_{lower} for θ , satisfying the constraint interval $\Theta = [\theta_{lower}, \theta_{upper}]$ that:

$$\theta_{upper} \geq \theta \geq \theta_{lower} \quad (4)$$

By iteratively identifying the minimal upper bound θ_{upper} and maximal lower bound θ_{lower} , we progressively narrow the interval Θ to converge on the optimal boundary θ_{opt} . The algorithm's core objective is to continually reduce θ_{upper} while increasing θ_{lower} , thereby shrinking the interval space until convergence.

1) *Initialization*: When a *Tiga* tag associates with a new cell, it initially operates in active mode to synchronize with the gateway (see §IV-D). The tag then sets a high initial boundary $\theta_{initial}$, ensuring successful communication whenever passive communication is feasible.

We then set $\theta_{upper} = \theta_{initial}$ and $\theta_{lower} = 0$, representing the upper and lower bounds of the boundary. Note that our mode selection criterion operates between the physical and MAC layers without affecting their protocol specifications. Next, we introduce the boundary search process of the *Tiga* tag in detail.

2) *Phase 1: Slow Start*: The *Tiga* tag initiates operation in the slow start phase, functioning in active mode. The boundary θ begins to descend with a gradually increasing rate of decline. The slow start phase ends when the θ reaches the real-time RSSI, transitioning the system to the fluctuation avoidance phase. During Phase 1, θ is first adjusted cautiously to prevent oscillations near the real boundary. The progressive acceleration of the descent rate enables rapid detection of the actual boundary, thereby maximizing opportunities for passive communication.

3) *Phase 2: Fluctuation Avoidance*: Upon reaching the real-time RSSI, θ rapidly decreases by $\Delta\theta$ and then remains fixed. Due to noise and variations in $L_1(d_i, t)$ (see §IV-A), RSSI fluctuations may occur. Without this phase, the *Tiga* tag would frequently switch between active and passive modes within short time intervals. Therefore, Phase 2 effectively mitigates "mode oscillation" of tags around real-time RSSI.

4) *Phase 3: Reliability Attempt*: Once θ is fixed, *Tiga* tag operates in passive mode and starts a timer to test reliability, encountering three possible situations, as illustrated in Fig. 7.

Algorithm 1: Blind Boundary Search Algorithm

Input: measured RSSI: $RSSI$, avoidance value: $\Delta\theta$, initial boundary: $\theta_{initial}$, termination value: $\Delta\theta_*$, received NAK: NAK

Output: optimal boundary: θ_{opt}

Initialize: $\theta_{upper} \leftarrow \theta_{initial}$, $\theta_{lower} \leftarrow 0$, $\theta \leftarrow \theta_{initial}$, state $\leftarrow PHASE_1$;

while $|\theta_{upper} - \theta_{lower}| > \Delta\theta_*$ **do**

get $RSSI$, NAK ;

switch state **do**

case $PHASE_1$ **do**

if $\theta > RSSI$ **then**

$\theta \leftarrow \max(\theta - rate, \theta_{lower})$;

else

state $\leftarrow PHASE_2$;

case $PHASE_2$ **do**

$\theta \leftarrow \max(\theta - \Delta\theta, \theta_{lower})$;

state $\leftarrow PHASE_3$;

case $PHASE_3$ **do**

Start Timer T ;

if No NAK & $RSSI > \theta$ during T **then**

$\theta_{upper} \leftarrow \theta$;

else if $RSSI < \theta$ during T **then**

state $\leftarrow PHASE_1$;

else if NAK **then**

$\theta_{lower} \leftarrow \theta$;

state $\leftarrow PHASE_4$;

case $PHASE_4$ **do**

$\theta \leftarrow \theta_{upper}$;

state $\leftarrow PHASE_1$;

$\theta_{opt} \leftarrow \theta_{upper}$;

- 1. If *Tiga* tag remains stable in passive mode during the timing period and does not receive an NAK from the gateway, it updates the upper bound to $\theta_{upper} = \theta$.
- 2. If the RSSI drops below the boundary θ , the tag switches to active mode and re-executes Phase 1 using the current boundary as the starting point.
- 3. If the gateway fails to receive a packet within the timeout period, it sends an NAK. Upon receiving the NAK, *Tiga* tag updates the lower bound to $\theta_{lower} = \theta$.

The gateway maintains a local timer that resets upon successfully receiving a packet from the tag. If the gateway does not receive a passive communication packet for an extended period, it indicates that the boundary selected by the *Tiga* tag is below the actual boundary. To implement the NAK, the gateway turns off the carrier signal, creating a steep falling edge that the tag can detect.

5) *Phase 4: Fast Recovery*: If *Tiga* tag receives an NAK, the boundary quickly recovers to the upper bound θ_{upper} and Phase 1 restarts from θ_{upper} .

Upon completing these four phases, the *Tiga* tag iteratively narrows the upper and lower bounds of the boundary through

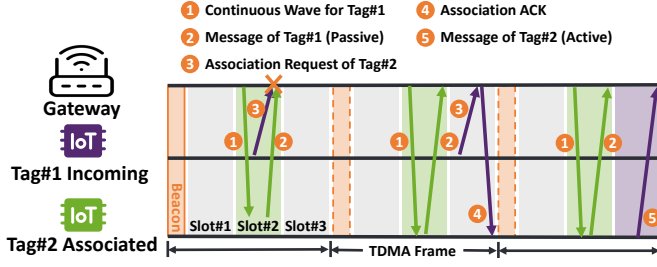


Fig. 8. *Tiga* association process: tag#1 as a new-comer and tag#2 has been associated.

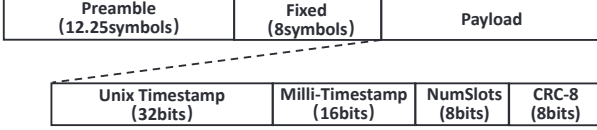


Fig. 9. Beacon frame structure.

an algorithmic process. The termination criterion for this algorithm is when the size of the constraint interval Θ becomes smaller than a termination value $\Delta\theta_*$, at which point the optimal boundary is determined as $\theta_{opt} = \theta_{upper}$. The value of parameter $\Delta\theta_*$ directly influences the deviation between the estimated boundary θ and the real boundary. Selecting an excessively small $\Delta\theta_*$ may lead to slow convergence, making it difficult to satisfy the termination condition. Therefore, during the implementation of the practical prototype system, the choice of a should be carefully determined based on the quantization precision of the selected ADC hardware. In our prototype design (See §V), we employ an 12-bit ADC with a quantization range of $[0, 4095]$, and set $\Delta\theta_* = 10$.

The blind boundary search algorithm (Algorithm 1) is designed to converge on the real boundary by continuously updating the upper and lower bounds based on the RSSI. Its effectiveness depends on environmental conditions and the tag's movement. If strong interference signals are present or the tag continues to receive NAKs after the algorithm converges, the final boundary is deemed invalid, and the search process must be restarted.

D. Protocol design

This section presents the design of the access protocol, as depicted in Fig. 8. The prototype system implements a TDMA-based scheme that allocates distinct time slots to individual tags, effectively mitigating interference.

Initially, tags operate in active mode to monitor the network. Each TDMA frame preamble contains a beacon field. The gateway periodically broadcasts beacons comprising a millisecond-level timestamp for synchronization and the total number of slots for transmissions (Fig. 9). Upon beacon reception, each tag randomly selects a time slot to transmit its association request. The gateway responds with an ACK upon successfully receiving a request in an idle slot, thereby establishing association. In collision scenarios, no association ACK is transmitted. Tags initiating unacknowledged requests subsequently execute random backoff procedures. Following this, each tag performs a boundary search within its allocated

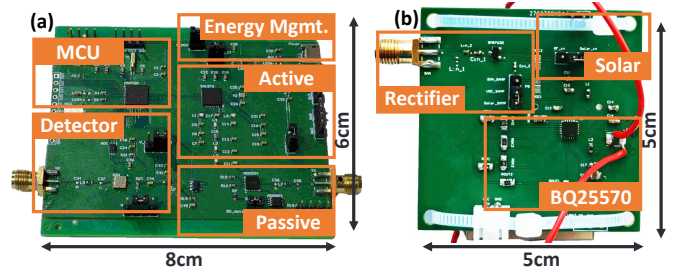


Fig. 10. *Tiga* tag hardware prototype. (a) Core board. (b) Energy harvesting board.

time window before transitioning to hybrid active-passive mode.

An example of this process, involving two tags and three time slots, is illustrated in Fig.8. Tag #2 is already associated in Slot #2 and operates in passive mode. Tag #1, as a new-comer, randomly sends an association request during Slot #2. However, the gateway fails to receive the request during the allocated time slot. Consequently, Tag #1 does not receive an association ACK and performs a random backoff before retransmitting. After the random backoff, Tag #1 resends the request in Slot #3. This time, the gateway successfully receives the request and sends the association ACK. Our MAC layer design is scalable to multi-tag scenarios, as demonstrated in §VI-F, where the evaluation shows that the number of concurrent tags supported by *Tiga* is four.

V. IMPLEMENTATION

The off-the-shelf *Tiga* prototype consists of two parts: a tag and a gateway. The implementation details are as follows. Although our general target is to propose a hybrid active-passive communication mechanism, we design a prototype supporting the LoRa PHY for deployment and evaluation.

***Tiga* tag Hardware.** The *Tiga* tag prototype consists of a core board and an energy harvesting board. The core board is prototyped on a four-layer PCB using commercial off-the-shelf circuit components as shown in Fig. 10(a). It consists of five key parts: the active communication module, the passive communication module, energy management, the dual-function low-power detector, and the MCU.

The generic baseband and control state machine are written in C and deployed on a low-power MCU, MSP430FR5969[75]. The energy management system, with regulators controlled by the MCU, powers the remaining circuit. The active communication module, SX1276IMLRT[76], supports communication at center frequencies of sub-GHz. The passive communication module uses the ADG902 RF switch[77], controlled by the MCU for backscatter. We use MSP430FR5969 to generate a 16MHz SMCLK signal as a frequency shift signal and pass it through XOR with the baseband signal. An ADG904 RF switch[78] is used for mode selection. The dual-function low-power detector consists of a passive demodulation circuit and a low-speed 12-bit ADC, ADS7042[79]. Since the antenna is single-ended, it is easier to achieve high passive voltage gain with a single-ended matching network and SAW filter TA0692A[80]. The envelope detector is built using SMS7630 [81] with 0.3V on-state voltage.

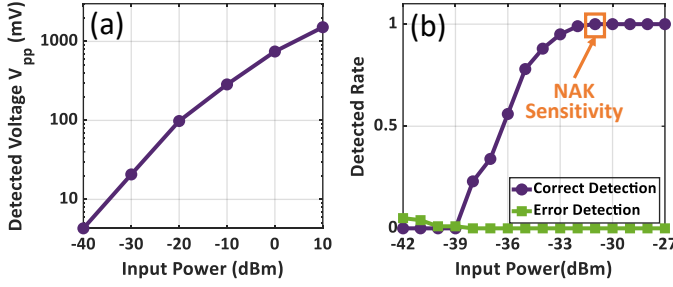


Fig. 11. Effectiveness of low power detector. (a) The detected voltage of RSSI detection. (b) The NAK sensitivity of detector.

As shown in Fig. 10(b), designed as a two-layer PCB, the energy harvesting board is dedicated to energy harvesting. We implement a one-stage rectifier for RF energy harvesting, comprising two Schottky diodes [81], two capacitors, and an impedance-matching circuit. The rectifier's output is connected to a BQ25570 power management IC, using 90mF AVX-BestCap[82] to store energy. Furthermore, interfaces for additional energy sources (e.g., solar) are reserved to accommodate user-configurable options.

Gateway. The gateway comprises two commercial LoRa RF Modules, SX1276RF1KAS[83], connected to the TI MSP-EXP430FR5969 LaunchPad[84]. It operates in frequency division duplexing, similar to the commercial LoRa gateway E870-L470LG11[85]. These two modules function as a transmitter and a receiver, respectively. The system can operate as a gateway by configuring the SPI interface and timer.

VI. EVALUATION

A. Experimental setup

In the subsequent evaluation, we experiment with the following parameter configurations. The gateway operates at 470MHz and 486MHz for uplink and downlink, respectively. The payload of each LoRa packet consists of 20 symbols. We use the *Tiga* tag to realize XORLoRa[86]. The default spreading factor (SF), coding rate (CR), and bandwidth (BW) of the LoRa signal are 8, 4/6, and 250kHz, respectively. The transmission power of the gateway is 14dBm. We use omnidirectional suction cup antennas with 3dBi gain for all experiments. We configure the reference voltage of the ADC to be 1.8V. The ADC quantizes the sampled value from 0 to 4095 and sends it to the MCU via the SPI bus. The RF rectifier of the energy harvesting board is configured at 915 MHz. In the actual experimental scenario, there are no strong interference signals in the uplink, downlink and energy harvesting operating bands.

B. Power Consumption

Prototype power consumption. *Tiga* tag uses the excess energy of the gateway to conserve its own energy. We systematically analyze the power consumption of the prototype by supplying a constant 3 V and using the Otii Arc energy analyzer [87]. The power consumption, detailed in Table. III, includes contributions from the RF radio, RF switch, low-speed ADC, regulators, and MCU. Under active operation, the total power consumption reaches 90.12mW for transmission (TX)

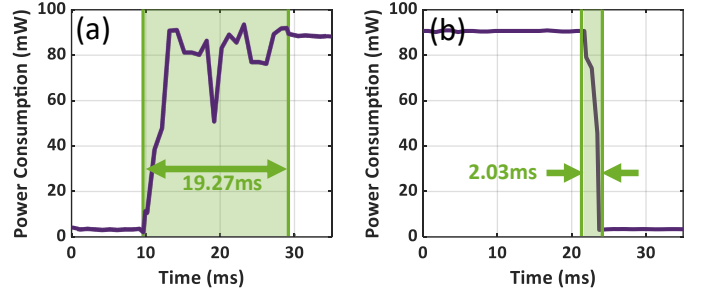


Fig. 12. Hardware cold-start latency. (a) Latency from passive to active mode. (b) Latency from active to passive mode.

TABLE III
POWER CONSUMPTION OF EACH COMPONENT IN *Tiga*

Components	Power Consumption (mW)
RF Radio (Active Mode TX/RX)	87.22 / 36.16
Packet Detection (Passive Mode RX)	~0
Crystal&RF switches (Passive Mode TX)	0.08
Low-Speed ADC	0.06
Modulation and Controller (MCU)	2.73
Power Management (Regulators)	0.03
Active Mode (TX / RX)	90.12 / 39.06
Passive Mode (TX / RX)	2.90 / 2.82
Sleep Mode	0.028

and 39.06mW for reception (RX), with the RF transceiver accounting for 96.78% of the energy budget. Conversely, by deactivating the high-power RF components, the passive mode curtails power consumption to a mere 2.90mW.

Sleep mode. We further measure the power consumption of sleep mode, where the MCU operates at LPM3.5, achieving 28μW. In this mode, both the RF front-end and other peripheral components of the *Tiga* tag are completely powered off, while the MCU maintains only a wake-up timer. This ultra-low-power configuration enables the *Tiga* tag to operate in an extremely power-efficient duty-cycled manner.

C. Micro-benchmarks

1) Effectiveness of low power detector:

(1) RSSI detection. We measure the output power of the envelop detector under different transmission power levels via a wired connection. A signal generator N5171B serves as the 486MHz RF transmitter, which precisely controls the output power level. The *Tiga* tag is directly connected to the transmitter via a coaxial cable. Since we employ ADC sampling for RSSI acquisition, the detected voltage V_{pp} from the envelope detector directly determines the RSSI value. Using an oscilloscope DSOX3054T, we measure V_{pp} , with results presented in Fig. 11(a). Our evaluation shows that as the transmission power increases from -40dBm to 10dBm, the output V_{pp} ranges from 4mV to 1213mV. The *Tiga* tag hardware incorporates a 12-bit ADC with a reference voltage set to 1.8V, enabling a detection resolution of $\frac{1800mW}{4096} = 0.44mV$.

(2) NAK detection. During the operation of *Tiga*'s boundary search algorithm, the system must detect NAK from the gateway. The NAK implementation involves the gateway abruptly terminating its carrier signal, creating a steep energy drop to a low level. We employ a sliding-window filter on the MSP430 to remove high-frequency noise from the sampled

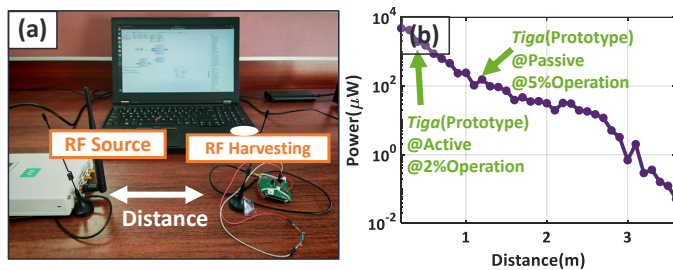


Fig. 13. Experiments for RF energy harvesting. (a) Experimental setup. (b) Curve of harvesting energy as a function of distance.

signal and extract the gradient of the RSSI sequence to track the NAK. A correct detection occurs when the *Tiga* tag successfully identifies the steep RSSI drop to a low level, while an error detection denotes false identification of this RSSI transition. NAK sensitivity is defined as the input power level where the NAK detection rate exceeds 99%. The detection results, illustrated in Fig. 11(b), demonstrate our detector achieves an NAK sensitivity of -31dBm. However, when input power falls below -39dBm, the detector has difficulty detecting NAK, while missed detections occur below -31dBm.

The NAK sensitivity is constrained by the detector's voltage output and the ADC's sampling resolution. Fig. 11(a) shows that when the input power is below -31dBm, the detector's output voltage drops below 20mV, and when the input power is below -39dBm, the output voltage falls below 5mV. Considering that the quantization step of the 12-bit ADC is 0.44mV and the RSSI fluctuation, it is challenging to detect the abrupt gradient drop of the RSSI signal in the MCU.

2) *Response to mode switching*: *Tiga*'s operational mechanism enables autonomous switching between active and passive modes. Here, we evaluate the response latency during mode switching, which is primarily influenced by two factors: control switching latency and hardware cold-start latency.

(1) **Control switching latency**. The mode switching process is governed by the MSP430 microcontroller, which manages power domains and discrete components. The control switching latency exhibits two key characteristics: First, the sampling rate of the ADC exhibits a linear relationship with its power consumption. According to the datasheet[79], the ADS7042 consumes $69\mu W$ at 100kSPS and $690\mu W$ at 1MSPS with 3V. In practical deployments, since the channel RSSI remains stable within $10\mu s$, a 100kSPS sampling rate suffices for our requirements. Second, after acquiring sampled values, the MSP430 sequentially executes control statements to enable/disable relevant power domains and components. The 8MHz MCLK clock speed in MSP430 does not constitute the primary bottleneck for switching latency. Consequently, the control switching delay is predominantly determined by the ADC sampling rate, which remains below 1ms in our system.

(2) **Hardware cold-start latency**. In passive mode, the energy management module deactivates power domains associated with the SX1276 transceiver. Thus, transitions from passive to active mode must account for the hardware latency induced by SX1276 cold-start. We use the Otii Arc energy analyzer [87] to record power consumption profiles during mode switching, as illustrated in Fig. 12. Experimental results

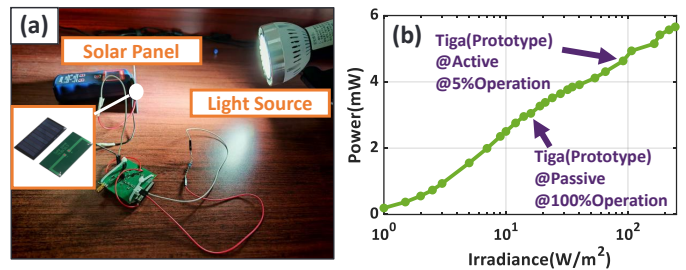


Fig. 14. Experiments for solar/light energy harvesting. (a) Experimental setup. (b) Curve of harvesting energy as a function of irradiance.

indicate that activating the RF front-end power domains and initializing the SX1276 requires $19.27ms$, whereas deactivating active components during transitions to passive mode incurs a latency of only $2.03ms$. These measurements confirm that hardware cold-start is the dominant factor limiting *Tiga*'s mode-switching responsiveness.

3) *RF energy harvesting*: In contrast to the uplink/downlink frequencies, we employ a 915MHz signal for energy harvesting. Using a USRP B210 equipped with an MWPA0050G04 RF amplifier and a 3dBi antenna, a transmit power of 36dBm is achieved at 915MHz. As shown in Fig. 13, we vary the distance between the *Tiga* energy harvesting board and the USRP to measure the harvested power, with a $10k\Omega$ load matching the input impedance specification of the BQ25570. The *Tiga* prototype's single-stage rectifier generates sufficient power to support active operation (2% duty cycle) at a 0.4-meter distance, while passive operation (5% duty cycle) is maintained at 1.2 meters. The results indicate that RF energy alone struggles to sustain prolonged active communication when the RF energy source is far away. However, after ASCII optimization[53], [54], [55], [56], the passive communication achieves μW -level power consumption, enabling long-term operation powered by RF energy.

4) *Solar energy harvesting*: To evaluate solar energy availability for *Tiga* tag, we set up an indoor experiment using a variable-power LED as the light source. A $53mm \times 30mm$ mini solar panel [88] is employed for energy harvesting, while a solar power meter SM206E-Solar [89] monitors the irradiance level. As shown in the Fig 14, when irradiance reaches $14W/m^2$, *Tiga* tag can operate in passive mode all the time. When irradiance is increased to $91W/m^2$, *Tiga* tag can operate in active mode with an 5% duty cycle. Since sunlight is attenuated by the atmosphere, the maximum normal surface irradiance at sea level on a clear day is approximately $1000W/m^2$ [90]. Therefore, outdoor deployment of *Tiga* tag can readily support its operation.

In §VI-G, we evaluate the real-world performance of the *Tiga* tag working in a hybrid active-passive mode in an indoor and outdoor scenarios.

D. Optimal boundary search

Tiga can search for the minimum upper bound and maximum lower bound of the boundary, thereby continuously narrowing the constrained interval Θ to obtain the optimal boundary θ_{opt} . Below, we evaluate the performance of *Tiga* in the actual optimal boundary search process.

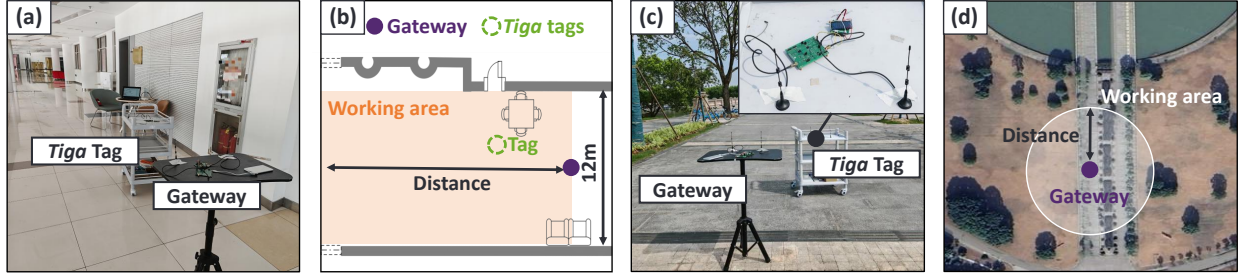


Fig. 15. Indoor and outdoor experiment setup. (a) Indoor scenario. (b) Indoor floor plan. (c) Outdoor scenario. (d) Outdoor floor plan.

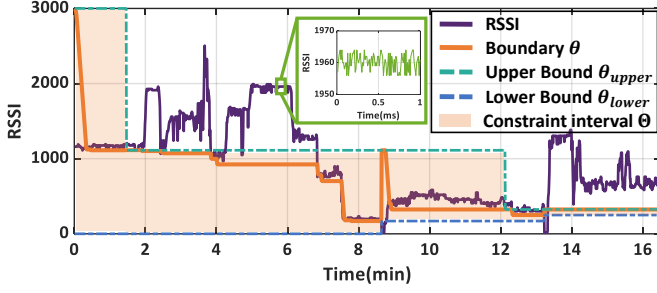


Fig. 16. Single TIGA tag performs Blind Boundary Search Algorithm in indoor scenes.

Setup. We conduct practical evaluations in both indoor and outdoor environments (as shown in Fig. 15 (a) and (b)), using a 3V battery as the energy source for the TIGA tag. Since the performance of the Blind Boundary Search Algorithm depends on dynamically varying RSSI, we deploy a TIGA tag within an unobstructed working area 30 meters from the gateway, allowing it to stop and move freely. This setup induces frequent RSSI fluctuations, simulating a scenario for elderly health monitoring. Under the experimental configuration described in §VI-A, Fig. 16 visually records the specific numerical changes of parameters during one such experiment.

1) *Convergence time:* We first evaluate the time required for TIGA to terminate the optimal boundary search algorithm, i.e., the convergence time. With a 12-bit ADC configuration, the TIGA tag is set with $\theta_{initial} = 3000$, $\Delta\theta_* = 10$ and $\Delta\theta = 10$. We conduct 50 experiments each in indoor and outdoor scenarios. As illustrated in Fig. 17(a), the convergence time primarily falls within the 10-30 minute range. In indoor environments, TIGA requires additional time to reach algorithm termination, attributed to the prevalence of complex multipath signal propagation.

2) *Impact of termination value:* Next, we vary the termination value $\Delta\theta_*$ from 6 to 16 while keeping $\theta_{initial} = 3000$ and $\Delta\theta = 10$. The convergence rate is defined as the ratio of successful optimal boundary search experiments completed within 40 minutes to the total number of experiments. The average time is defined as the mean duration of all successful experiments. We perform 50 experiments for different $\Delta\theta_*$ values in both indoor and outdoor scenarios. Fig. 17(b) shows that when $\Delta\theta_* > 10$, 98% of experiments successfully locate the optimal boundary in both environments. If $\Delta\theta_*$ is too small, the constrained interval Θ struggles to satisfy $\Theta < \Delta\theta_*$, prolonging convergence time. Conversely, a larger $\Delta\theta_*$ facilitates successful searches but increases the error relative to the actual boundary.

3) *Impact of avoidance value:* We vary the avoidance value $\Delta\theta$ from 6 to 16 while keeping $\theta_{initial} = 3000$ and $\Delta\theta_* = 10$ constant. Similarly, 50 experiments are conducted for different $\Delta\theta$ values in both scenarios. Fig. 17(c) demonstrates that 98% of experiments successfully identify the optimal boundary in both environments. Additionally, subplots in Fig. 16 reveal that RSSI fluctuates within a range of less than 10 over 1ms. Thus, a smaller $\Delta\theta$ causes the MCU to frequently switch between active and passive modes, yet the optimal boundary can still be found after multiple Phase 2 iterations. However, a larger $\Delta\theta$ may delay termination opportunities, increasing convergence time.

In subsequent experiments, the TIGA tag is configured with $\theta_{initial} = 3000$, $\Delta\theta_* = 10$ and $\Delta\theta = 10$.

E. Head-to-head comparison

Now we compare the performance of a single TIGA tag with active LoRa implemented on SX1276 and XORLoRa[86], using a 3V battery as the power source. The experiments are conducted in an indoor environment, as shown in Fig. 15(a). We deploy the tags along a straight line at distances ranging from 5 to 30 meters from the gateway, moving back and forth at a speed of 0.2m/s for five rounds, with LoRa packet transmissions completed every 500ms. In the active LoRa tests, the gateway served solely as a receiver, whereas in the XORLoRa and TIGA experiments, the gateway functioned as both a receiver and an excitation signal source. For each distance setting, we conduct 50 experiments. We compare the three schemes in terms of goodput, power consumption, and power efficiency, with the results shown in Fig. 18.

Goodput. The effective PHY goodput is computed by multiplying the transmitted bit rate by the fraction of successfully decoded packets. Fig. 18(a) demonstrates that active LoRa on SX1276 achieves near-full-capacity data transmission stably, whereas XORLoRa, which employs passive communication, exhibits a decline in correctly received and decoded packets as distance increases. While TIGA's goodput is marginally lower than that of active LoRa, it significantly outperforms passive LoRa at extended distances. The slight reduction compared to the active baseline is attributed to packet loss occurring during the optimal boundary search process.

Power consumption. We record the average power consumption for each experiment. Active LoRa switches to receive mode after completing a packet transmission. Fig. 18(b) shows that the power consumption of active LoRa and XORLoRa remains stable across different distances, whereas TIGA's

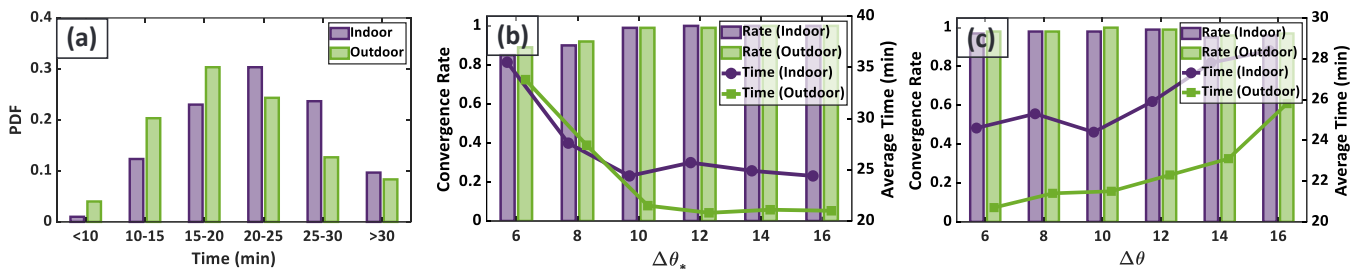


Fig. 17. Performance of optimal boundary search. (a) Distribution of convergence times. (b) Impact of termination value. (c) Impact of avoidance value.

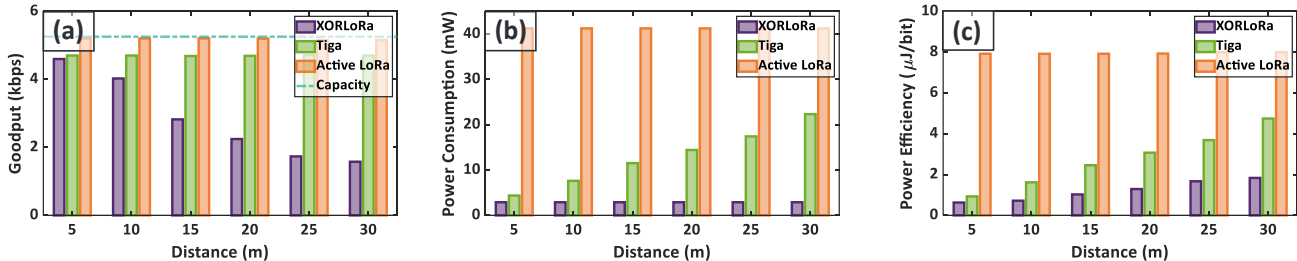


Fig. 18. Comparison of performance between active LoRa, *TIGA* and XORLoRa. (a) Goodput. (b) Power consumption. (c) Power efficiency.

power consumption increases with distance due to greater involvement of the active mode. Compared to active LoRa, *TIGA* reduces dynamic power consumption by 1.85 $\times$ to 9.42 $\times$ at distances of 5–30 meters from the gateway.

Power efficiency. Based on the measured goodput and power consumption, we calculate power efficiency, which refers to the energy required to transmit one bit in each experimental cycle. Note that a higher power efficiency value indicates greater energy consumption per transmitted bit. As shown in Fig. 18(c), both XORLoRa and *TIGA* exhibit increasing power efficiency with distance. For XORLoRa, although power consumption remains unchanged at long distances, the significant drop in goodput leads to higher power efficiency. For *TIGA*, the increased participation of the active mode in transmission rounds at longer distances results in higher power consumption, thereby increasing power efficiency.

F. Multi-tag scenario

To evaluate system performance under multi-tag coexistence, we deploy four *TIGA* tags in an indoor corridor environment (Fig. 15(a) and (b)) utilizing our proposed access protocol. Each tag is positioned within a 5-meter line-of-sight range of the gateway, supporting both stationary and mobile scenarios. The gateway is configured with a TDMA frame duration of 500ms and transmits synchronization beacons at one-minute intervals. Fig. 19(a) presents the goodput and power consumption for each tag, derived from 50 experimental trials.

Compared with single-tag performance (detailed in §VI-E), the per-tag goodput remained stable at approximately 4.6kbps, while the power consumption in the multi-tag scenario averaged 6.8mW. Each tag operates reliably within its allocated time slot. The observed increase in average power consumption, relative to the single-tag case, is primarily attributed to the additional energy overhead incurred during network protocol execution. Although this experiment is limited to four tags, we believe our system can scale to support more tags.

Additionally, we conduct MATLAB simulations to evaluate the access performance with 8 to 128 tags. The system is configured with 128 slots for tag access. The simulation ran for 10 seconds, with each 500ms TDMA frame. We assume all tags are initially synchronized, and each tag attempts access in every frame. We define the success rate as the ratio of successfully accessed tags to the total number of tags, while the collision rate represents the ratio of tags that failed to access due to contention. As shown in Fig. 19(b), our access protocol achieved over 95% success rate when the number of tags is below 80. However, as the tag count approached the slot capacity limit, the collision rate increased significantly.

G. Ambient IoT deployment

Here we evaluate the performance of *TIGA* in Ambient IoT scenarios. The gateway-to-tag distance is fixed at 10 meters, with channel conditions artificially affected using obstacle occlusion. We first deploy *TIGA* tag in a corridor (as shown in Fig. 15(a) and (b)) using a USRP B210 equipped with an MHPA0050G04 RF amplifier and a 3dBi antenna. The distance between the tags and USRP B210 is set to 0.1m, and we measure the output voltage of the energy harvesting board during one harvesting round. When the USRP B210 generates 36dBm output power, the purple curve in Fig. 20(a) records the voltage variation during one operational cycle of *TIGA*, showing 42s charging time enabling 73s operation. With 30dBm output power from USRP B210, the green curve depicts the voltage profile of active LoRa, requiring 135s charging for merely 18s operation. Furthermore, by employing different transmit powers for the energy source in the two experiments, it is observed that the received signal power at the tag significantly impacts the charging rate. Notably, when the charging rate fails to offset the static power consumption of the hardware, it becomes difficult for the tag to effectively harvest and accumulate energy.

Furthermore, we deploy *TIGA* in an outdoor environment (as shown in Fig. 15(c) and (d)) to evaluate its solar energy har-

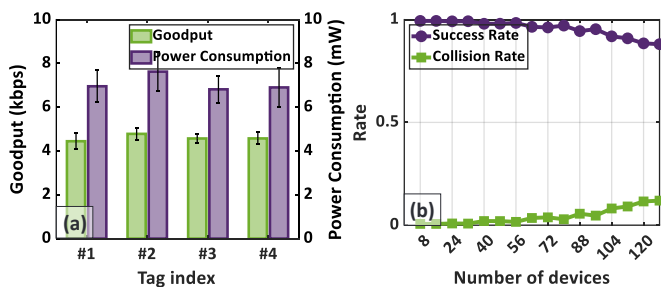


Fig. 19. Performance of *Tiga* in multi-tag scenario.

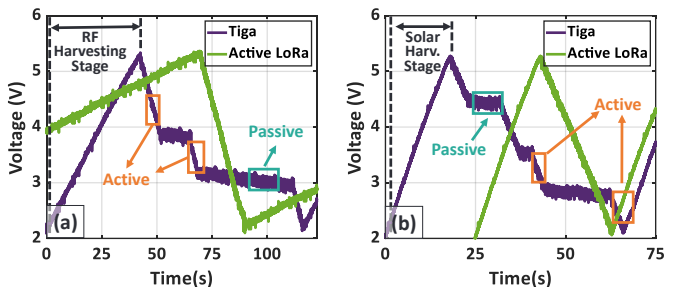


Fig. 20. During one harvesting round, the operating voltage comparison between *Tiga* and active LoRa.

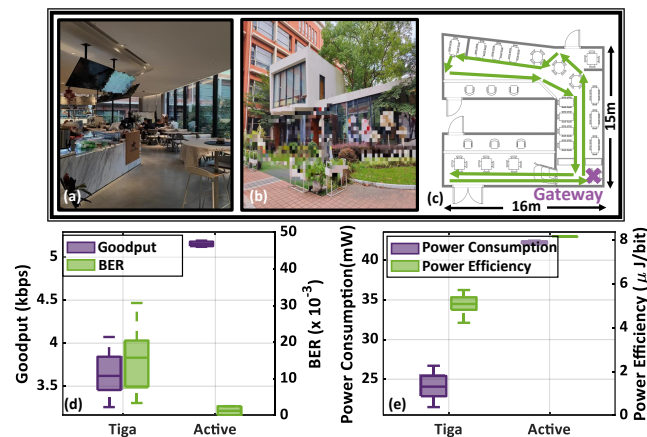


Fig. 21. Evaluation of *Tiga* in a detached house.

vesting performance under $317W/m^2$ irradiance. The purple curve in Fig. 20(b) shows that voltage drops sharply during active mode but declines gradually in passive mode. After 37 seconds of alternating between active and passive modes, *Tiga* re-entered the charging state, with parameter values from the *blind boundary search algorithm* stored in the MSP430's non-volatile FRAM. For comparison, we deploy an active LoRa node with the same scenario. The green curve demonstrates that with the same 18-second charging duration, active LoRa operates for only 17 seconds, whereas *Tiga* tag achieves $2.1\times$ longer operational time.

These results demonstrate that energy source power significantly impacts harvesting duration, where lower energy leads to longer harvesting time. Moreover, *Tiga* substantially extends operational time per harvesting cycle for A-IoT devices without compromising communication link reliability.

H. *Tiga* in action

To evaluate the efficiency and robustness of *Tiga* in real-world scenarios characterized by multipath effects and obstructions, we conduct a comprehensive case study in a detached house. This environment, depicted in Fig. 21(a) and (b), simulates a typical elderly health monitoring application where the subject moves between rooms.

Setup. A *Tiga* tag is attached to a human subject to monitor data. To simulate a realistic deployment, the gateway is positioned in a corner room on the second floor, while the subject traverses a complex path from the first floor, as illustrated in Fig. 21(c). The subject walks back and forth along this

route 20 times at a speed of approximately $1m/s$. We compare the performance of *Tiga* against a baseline fully active mode, conducting 50 independent trials for each scheme.

Result. The experimental results, presented in Fig. 21(d) and (e), confirm that *Tiga* effectively mitigates the reliability challenges of real-world environments while maintaining significant energy advantages. Although the fully active mode exhibits marginally higher goodput due to the superior penetration capability of high-power transmission, *Tiga* ensures continuous connectivity by autonomously switching to active mode when the blind boundary search algorithm detects RSSI drops caused by structural obstructions. This adaptive mechanism successfully mitigates the “dead zones” inherent to passive-only systems, allowing *Tiga* to achieve a $1.75\times$ reduction in power consumption compared to the baseline by opportunistically leveraging passive backscatter during favorable channel conditions.

VII. DISCUSSION AND FUTURE WORK

Compatibility of IoT protocols. We select LoRa PHY for deployment on *Tiga* as a hardware solution to enable ubiquitous connectivity. While LoRa PHY-enabled *Tiga* achieves significant power savings in hybrid active-passive communication deployments, the current LoRaWAN protocol standard does not natively support *Tiga*'s operation. However, *Tiga* is designed to be independent of the specific uplink PHY implementation and aligns well with emerging standards like the 3GPP A-IoT specification [91], [92] and IEEE 802.11 AMP [93], [94], which officially establish hybrid frameworks for active and passive devices. Crucially, both standards utilize OFDM-synthesized OOK for downlink communication. This modulation scheme is physically compatible with *Tiga*'s dual-function low-power detector, allowing it to inherently demodulate control signals and monitor channel energy. Consequently, *Tiga* can be seamlessly integrated into these standardized architectures by adapting its active RF front-end for high-rate transmission while reusing the low-power detector for efficient, standard-compliant downlink listening.

Target Application Scenarios. As a pioneer in exploring hybrid active-passive prototype deployment, *Tiga* paves the way for enhancing typical A-IoT applications, such as elderly health monitoring and warehouse inventory management. In elderly health monitoring, *Tiga* addresses the critical trade-off between battery longevity and emergency responsiveness. By opportunistically switching to active mode, it mitigates the risk of communication outages during critical events (e.g.,

falls), thereby offering a potential solution to improve system reliability in dynamic environments. Similarly, in warehouse inventory scenarios, *Tiga* helps alleviate the “blind spot” issue inherent in passive systems. By utilizing active transmission for items in challenging channel conditions, *Tiga* significantly improves coverage robustness, demonstrating the capability of hybrid architectures to adapt to complex real-world deployments.

Scalability in Large-scale Networks. Real-world A-IoT scenarios often necessitate the deployment of tens to hundreds of tags within a single gateway's coverage. We validate the feasibility of the *Tiga* system using a four-tag hardware prototype and extended the evaluation via simulations to support up to 128 nodes, achieving a success rate exceeding 95% for up to 80 concurrent tags using a TDMA-based protocol. It is important to underscore that the primary contribution of *Tiga* lies in the architectural innovation of autonomous active-passive mode switching based on energy boundaries. The proposed microwatt-level energy detector and the blind boundary search algorithm are designed to be decoupled from the specific PHY layer implementation. Consequently, *Tiga*'s architecture can be seamlessly integrated with high-concurrency mechanisms, such as Orthogonal Frequency-Division Multiple Access (OFDMA) [13], [44], [53], in our future work. This integration will enable *Tiga* to better utilize channel spectrum and support massive connectivity without compromising the energy-efficiency benefits of the hybrid architecture.

VIII. CONCLUSION

We present *Tiga*, an efficient hybrid active-passive communication system for A-IoT deployments featuring autonomous mode switching. *Tiga* employs a holistic hardware-software co-design approach to determine the optimal mode selection boundary. We introduce a dual-function low-power detector circuit capable of demodulating OOK signals while computing RSSI. Furthermore, we propose a four-stage blind boundary search algorithm that achieves effective boundary determination with small power consumption and computational overhead. Comprehensive experiments demonstrate that *Tiga* reduces power consumption by $1.85\times$ to $9.42\times$ across varying distances from the gateway. We believe *Tiga* provides a more practical pathway for future low-power IoT system deployments targeting A-IoT applications.

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